CHAPTER 3: CAPACITANCE & ELECTROSTATIC ENERGY STORAGE

3.1 Charge Accumulation and Electrostatic Displacement

A capacitor consists of two isolated conductors separated by a dielectric medium or vacuum. When an external potential difference V is applied, charge carriers accumulate on opposing conductive boundaries without direct conduction across the gap. The accumulated charge Q is strictly governed by the geometric capacitance: Q = C * V.

3.2 Dielectric Polarization and Bound Charge

Introducing an insulating dielectric material between conductive plates causes microscopic molecular dipole alignment. This dielectric polarization produces internal bound surface charges that partially counteract the applied field, attenuating net electric field strength and multiplying overall capacitance by the dielectric constant kappa: C = kappa * epsilon_0 * (A / d).

3.3 Electrostatic Potential Energy Density

Charging a capacitor requires thermodynamic work to overcome Coulombic repulsion from previously accumulated charges. This energy is stored directly within the electrostatic field filling the dielectric volume: U = 0.5 * C * V^2 = 0.5 * Q^2 / C. The spatial energy density of the field is given by u = 0.5 * epsilon * E^2.

3.4 RC Transient Charging Dynamics & Time Constant

When placed in series with a resistive element R, charge accumulation is rate-limited by the current constraint. The potential across the capacitor relaxes exponentially toward asymptotic equilibrium according to V(t) = V_0 * (1 - exp(-t / (R * C))). The product tau = R * C defines the fundamental characteristic relaxation timescale.

3.5 Mechanical Dualities: Spring Compliance vs. Capacitance

Capacitive energy storage behaves identically to elastic energy storage in a mechanical spring, where mechanical force corresponds to voltage, physical displacement corresponds to charge, and mechanical compliance (1/k) corresponds to capacitance C. These mathematical dualities govern harmonic oscillation in LC and RLC networks.